

***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

**CS/EC/IT 19-200-0105B & CE/EE/ME/SE 19-200-0205A BASIC MECHANICAL ENGINEERING
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Write a note on Carnot cycle and derive its efficiency equation.
- (b) Explain about thermodynamic equilibrium and its conditions.
- (c) Compare MPFI and carburetted engines.
- (d) A certain machine works on reversed Carnot cycle between temperatures of -10°C and 27°C . Find its COP when working as:
 - (i) Refrigerator
 - (ii) Heat pump
- (e) How are hydraulic turbines classified?
- (f) Compare centrifugal pump with reciprocating pump.
- (g) Differentiate between Brazing and Soldering process. How these welding processes are different from arc welding?
- (h) How hardening heat treatment process is done? Which heat treatment is carried out after hardening process?

PART B

(4 × 12 = 48)

- II. (a) Write a note on different thermodynamic processes and represent that processes in a pV diagram. (5)
- (b) 1 kg of air at a pressure of 8 bar and a temperature of 60°C expands to a pressure of 1 bar according to the law $pV^{1.25}$ is constant. Then the system is executing an isochoric process to attain the initial pressure. Determine (7)
 - (i) Total change in internal energy
 - (ii) Total Work done
 - (iii) Total heat transferred
 Assume $R = 287 \text{ J/kg.K}$ and $c_v = 0.718 \text{ kJ/kg.K}$

OR

- III. (a) Differentiate between boiler mountings and accessories and state their functions. (5)
- (b) 5 kg of air at 2 bar pressure and 75°C temperature at state 1 is compressed polytropically to state 2 at pressure 7.5 bar, index of compression being 1.3. And after that the system is executing an isobaric process to state 3. It is then cooled at constant temperature to its original state 1. Find the net work done and heat transferred. Assume $R = 287 \text{ J/kg.K}$ and $c_v = 0.718 \text{ kJ/kg.K}$. (7)
- IV. (a) Derive air standard efficiency of a Diesel cycle with the help of a neat P-V and T-S diagram. (7)
- (b) Explain any one ignition system used in I.C engine with the help of a neat diagram. (5)

OR

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- V. (a) With the help of a neat figure explain summer and winter air conditioning system. (7)
(b) Explain simple vapour compression refrigeration system with the help of a neat figure. (5)
- VI. (a) With the help of a neat sketch explain the working of an impulse turbine. (8)
(b) Explain how the impeller of a centrifugal pump converts mechanical energy into hydraulic energy of liquid. (4)
- OR**
- VII. (a) Draw a general layout of a hydro-electric power plant using a reaction turbine and explain the functions of the following (6)
(i) Scroll casing
(ii) Runner
(iii) Draft tube
(b) Explain the working of nuclear power plant with a neat diagram. (6)
- VIII. (a) Explain the function of following elements of a sand casting mould (6)
(i) Pattern
(ii) Core
(iii) Riser
(iv) Runner
(v) Sprue
(vi) Pouring basin
(b) State any two applications of rolling process. Explain different rolling arrangements with figure. (6)
- OR**
- IX. (a) Derive an expression for finding ratio of belt tensions of a driven pulley rotating in clockwise direction. (8)
(b) It has been decided to change a crossed belt in to an open belt. The pulleys attached to the two parallel shafts are 900 mm and 1200 mm diameter and the distance between the shafts is 7.5 metres. State whether the length of the belt has to be changed? If, it is to be changed, then find the change in length. (4)
